⚠️ **TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE**

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — "SentinelFlow 500" — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation's compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the **document register** for the full explanation.

Clause: 5.6 — Software Integration and Testing · **Register:** Device Example Register

Document ID: SOP-07 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C
Status: SOP — template (Procedure not yet written for SentinelFlow 500)

> **Fictional worked example.** See SOP-01 for the disclaimer that applies to this whole document set, and the Device Example Register for the SentinelFlow 500 device description and what "template" status means here.

1. Purpose

This SOP defines how SentinelFlow 500's software items are integrated in a planned sequence and tested as they come together, per Clause 5.6.

2. Scope

Applies to integration of SentinelFlow 500's control software items (Class C — see SOP-01 §7.3). Covers Clause 5.6 in full — every sub-clause of 5.6 applies at Class B and C; there is no integration testing requirement at Class A, since Class A software is not required to be architecturally decomposed into items in the first place (see SOP-04).

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Software Development Lead	Plans and documents the integration sequence before integration begins
Test Lead	Executes integration tests against each interface's specification; runs regression tests after fixes
QA/RA	Confirms integration test records are retained as objective evidence

4. Definitions & Abbreviations

TERM	MEANING
Integration testing	Testing that confirms integrated software items behave as described in the architectural design, not merely that they run without crashing
Regression testing	Re-testing after a fix to confirm no previously-passing behaviour was broken

5. References

- IEC 62304:2006+AMD1:2015, §5.6
- **SOP-04 — Software Architecture**: the interface specifications integration tests are written against
- **SOP-13 — Software Problem Resolution Process**: anomalies found during integration testing are managed through this process

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.6.1	Integrate software units	Yes	(No standard-named output.)
5.6.2	Verify software integration	Yes	Records of the evidence that software units were integrated in accordance with the integration plan
5.6.3	Software integration testing	Yes	Documented results of testing the integrated software items
5.6.4	Software integration testing content	Yes	(No standard-named output.)
5.6.5	Evaluate software integration test procedures	Yes	(No standard-named output.)
5.6.6	Conduct regression tests	Yes	(No standard-named output.)
5.6.7	Integration test record contents	Yes	Integration test records: the pass/fail result and list of anomalies, sufficient records to repeat the test, and the identity of the tester
5.6.8	Use software problem resolution process	Yes	(No standard-named output.)

(Quoted directly from `applicability.json`, identical in content to the self-referential set's 07, whose Status is itself marked **Partial** — that project has no formally partitioned software items to integrate. This device-example set's version is a template for the same clause, not a completed alternative.)

7. Procedure

Not yet written for SentinelFlow 500 — see the Device Example Register for what "template" status means. Once complete, this section would give SentinelFlow 500's actual integration sequence — e.g. Sensor Monitoring integrated with Motor Control before Alarm Management is layered on top — and the interface tests run at each step.

5.6.1–5.6.2 — Integrate and verify integration

[Template.]

5.6.3–5.6.5 — Integration testing, its content, and procedure evaluation

[Template.]

5.6.6 — Regression testing

[Template.]

5.6.7–5.6.8 — Test record contents and problem resolution

[Template — anomalies found here would be logged per SOP-13.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
Integration plan	Software Development Lead	Project technical file
Integration test records, per step	Test Lead	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending